



AM Transmitters and Demodulators

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Basics of Modulators

High Level Modulators:

Modulation takes place in the final element of final stage where carrier is at maximum amplitude.

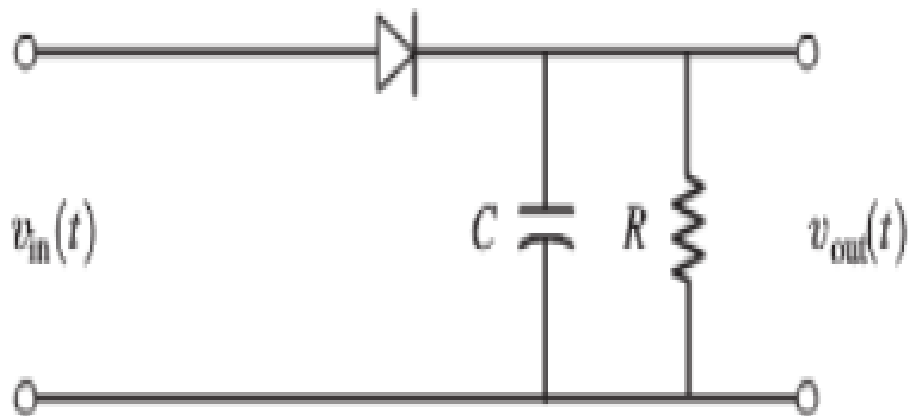
Low Level Modulators: Modulation takes place prior to the output element of the final stage of transmitter.

Methods of Detection

Non-Coherent Detection: No need to generate carrier at the receiver for demodulation

Coherent Detection: The carrier is locally generated at the receiver for demodulation.

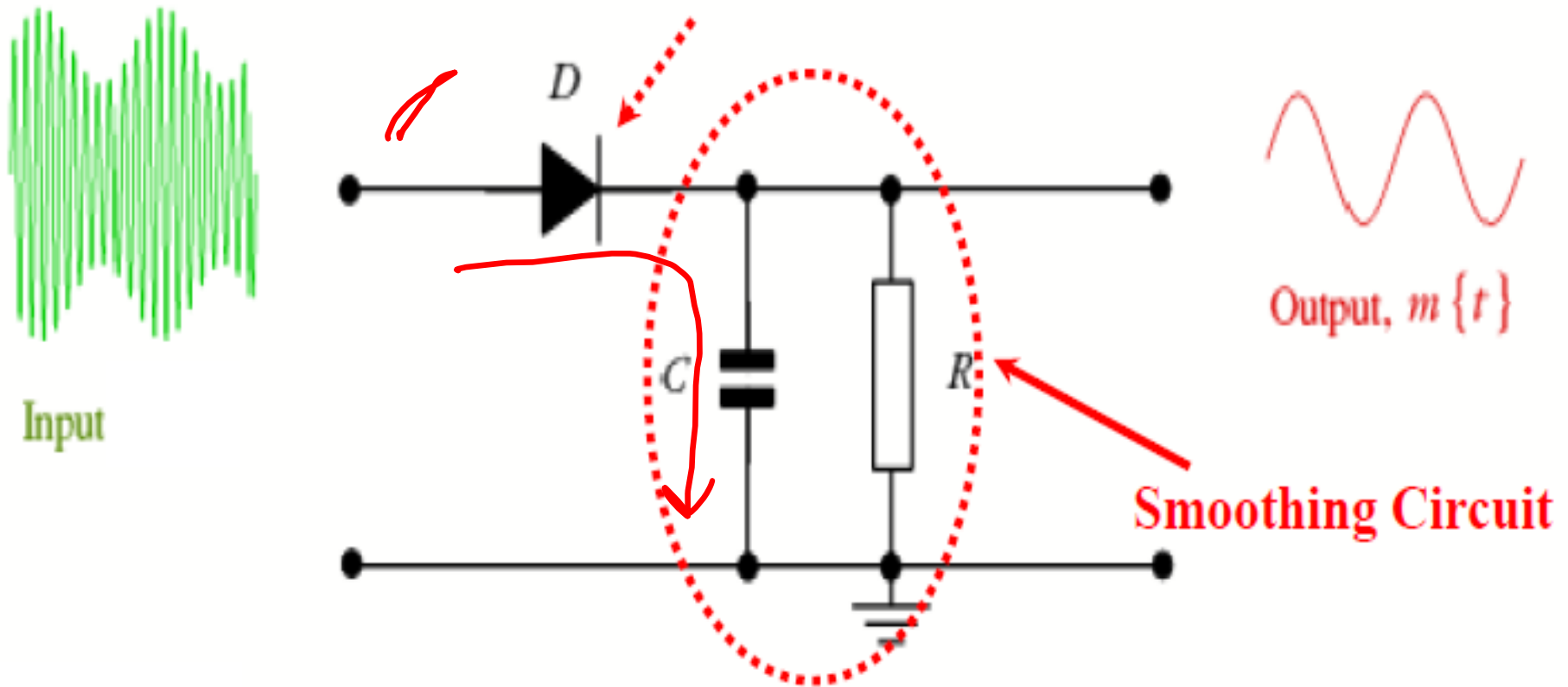
Simple Diode Detector



Basic operation: Assume $f_c = 300$ KHz and $f_m = 2$ KHz

- Then there will be frequencies 298, 300, 302 KHz
- The detector will detect many different frequencies
- AM frequencies + AM harmonics + SUM of AM frequencies + DIFF of AM frequencies
- The RC LPF is set to pass only DIFF frequencies

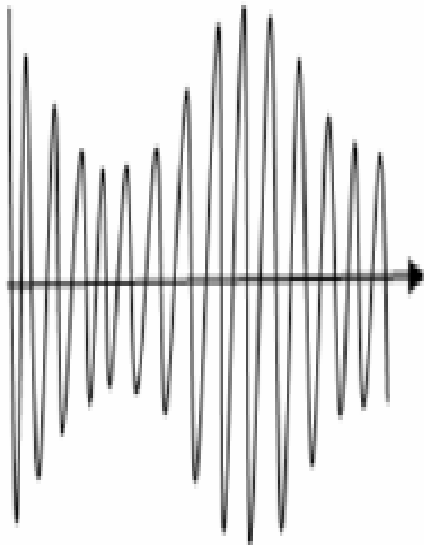
Circuit Diagram



Envelope Detector (AM Demodulator/Detector).

Waveforms at different stages

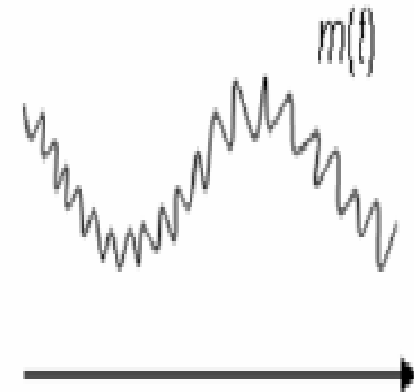
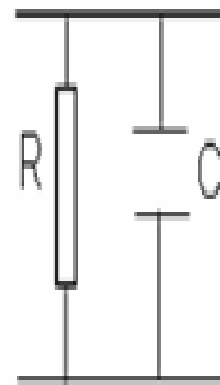
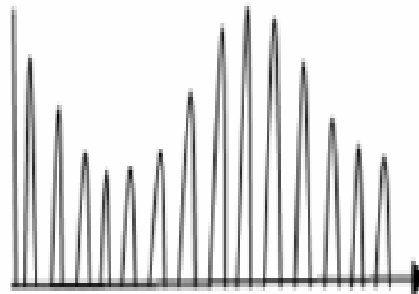
DSBAM



$m \leq 1$



Half wave rectified



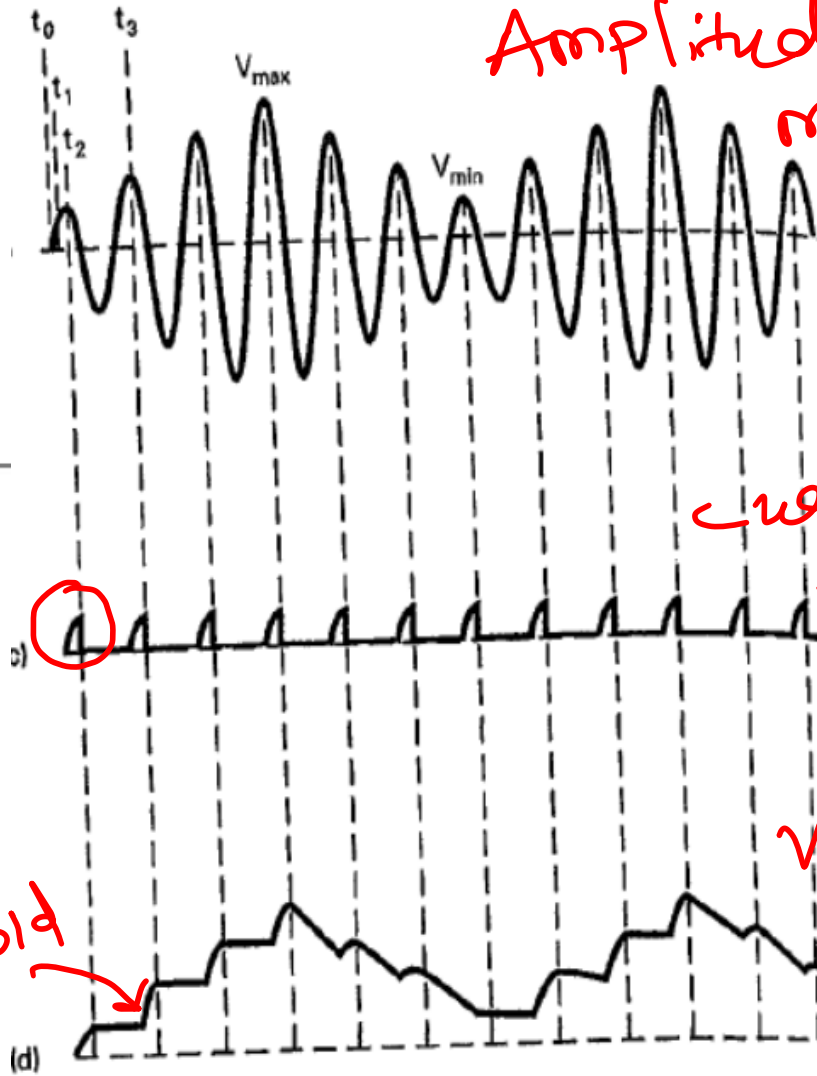
Smoothed
output

Simple Diode Detector Waveforms

Amplitude modulated

Input signal, $v_{in}(t)$

Output signal, $v_{out}(t)$



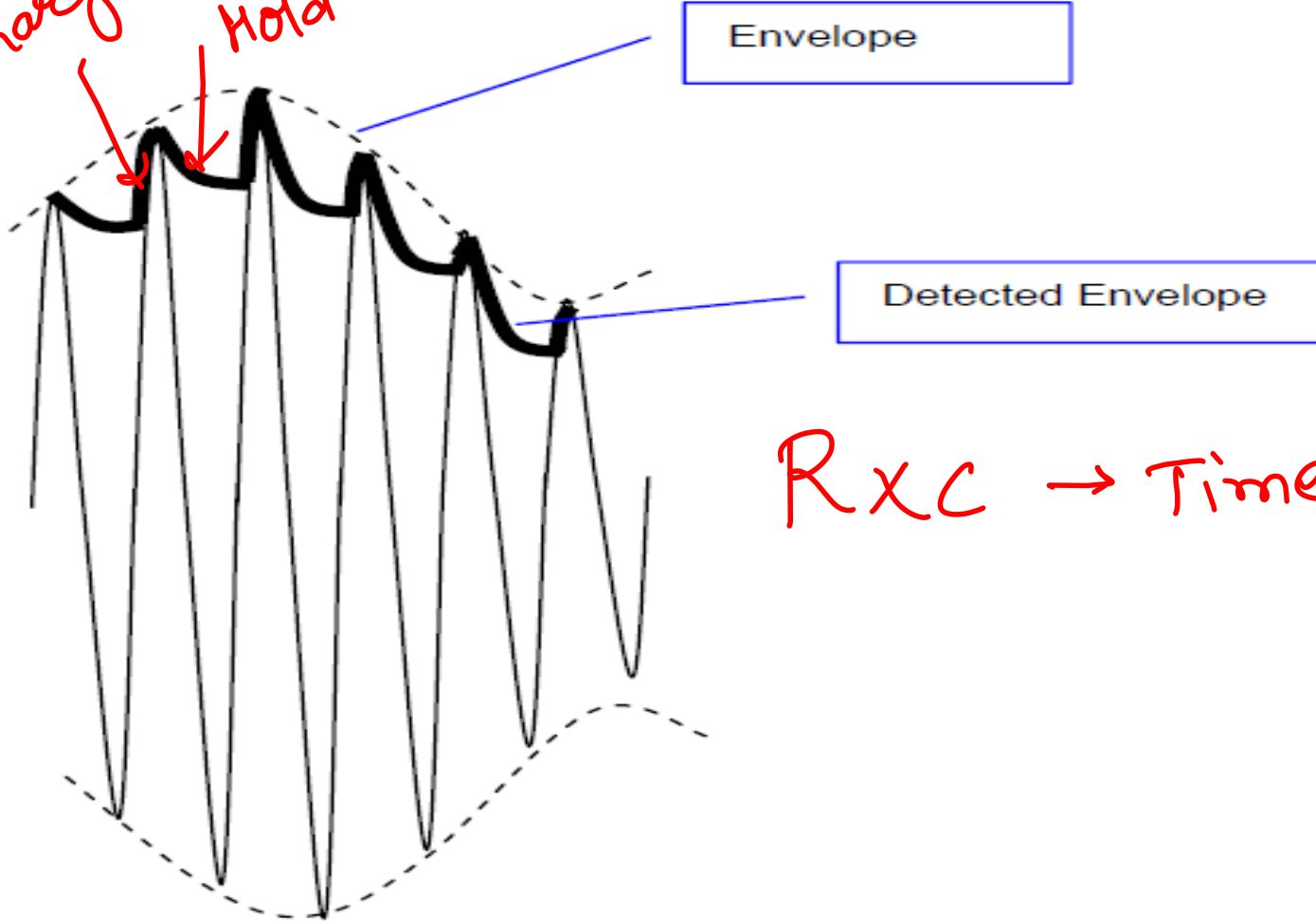
Current through diode

Voltage across capacitor

Hold

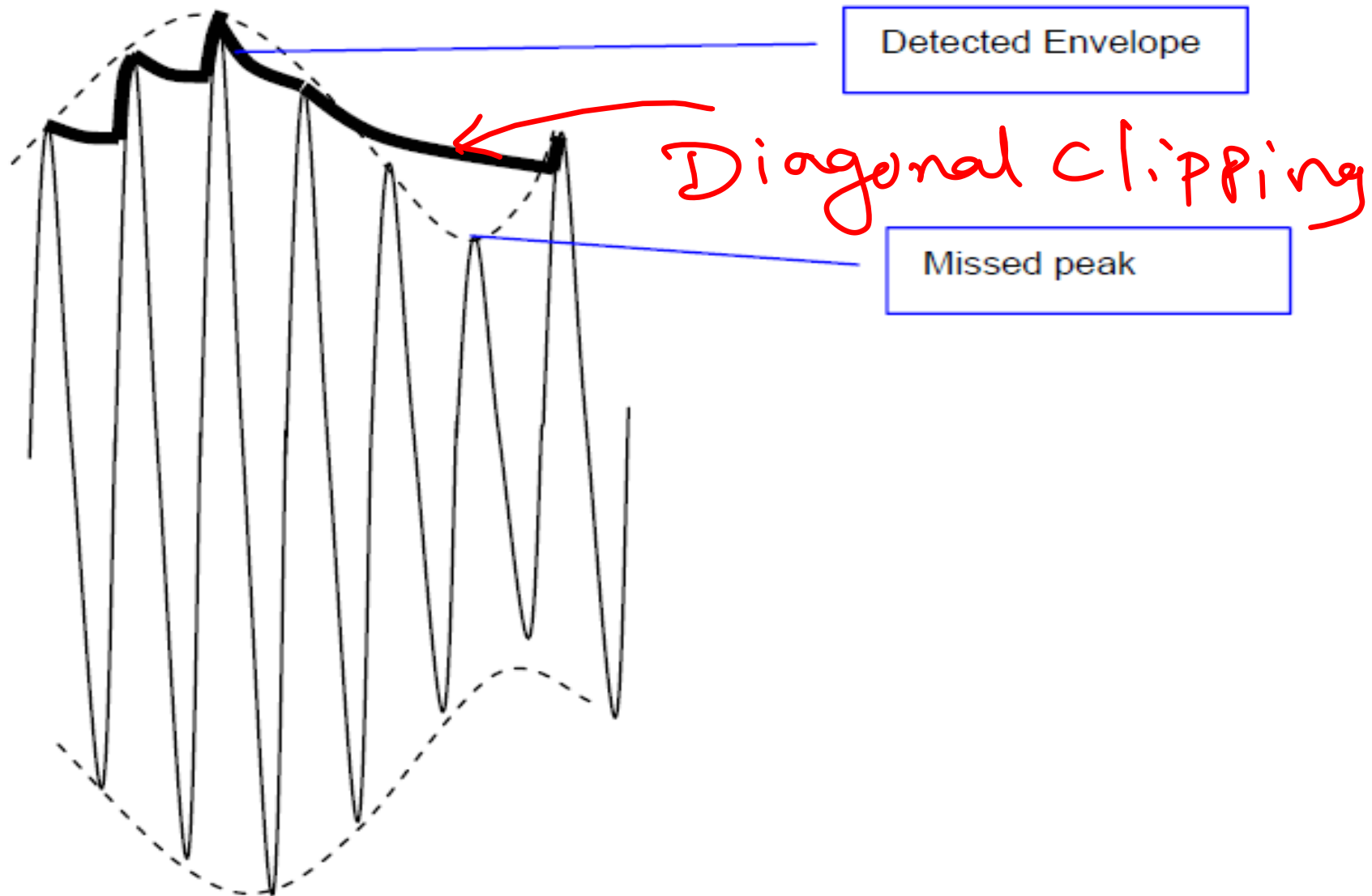
Waveforms

charging
hold



$RXC \rightarrow$ Time constant

Waveforms (RC time constant is large)



Waveforms (RC time constant is large and too small)

If RC is too small, the carrier wave produced at the output is shown in Fig. 4a.



Fig. 4a.

If RC is too large, the voltages or the signal at the output does not follow the envelope as is Fig. 4b.

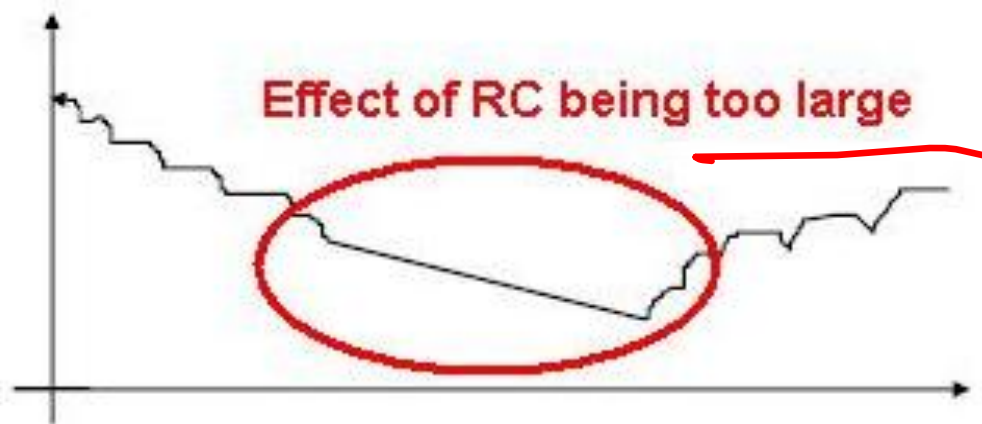


Fig. 4b

Diode Detector-Distortion

- What should be the value of RC?
 - If too low then discharges too fast
 - If too high the envelope will be distorted
 - The highest modulating signal:

$$f_{m(\max)} = \frac{\sqrt{(1/m^2) - 1}}{2\pi RC}$$

